

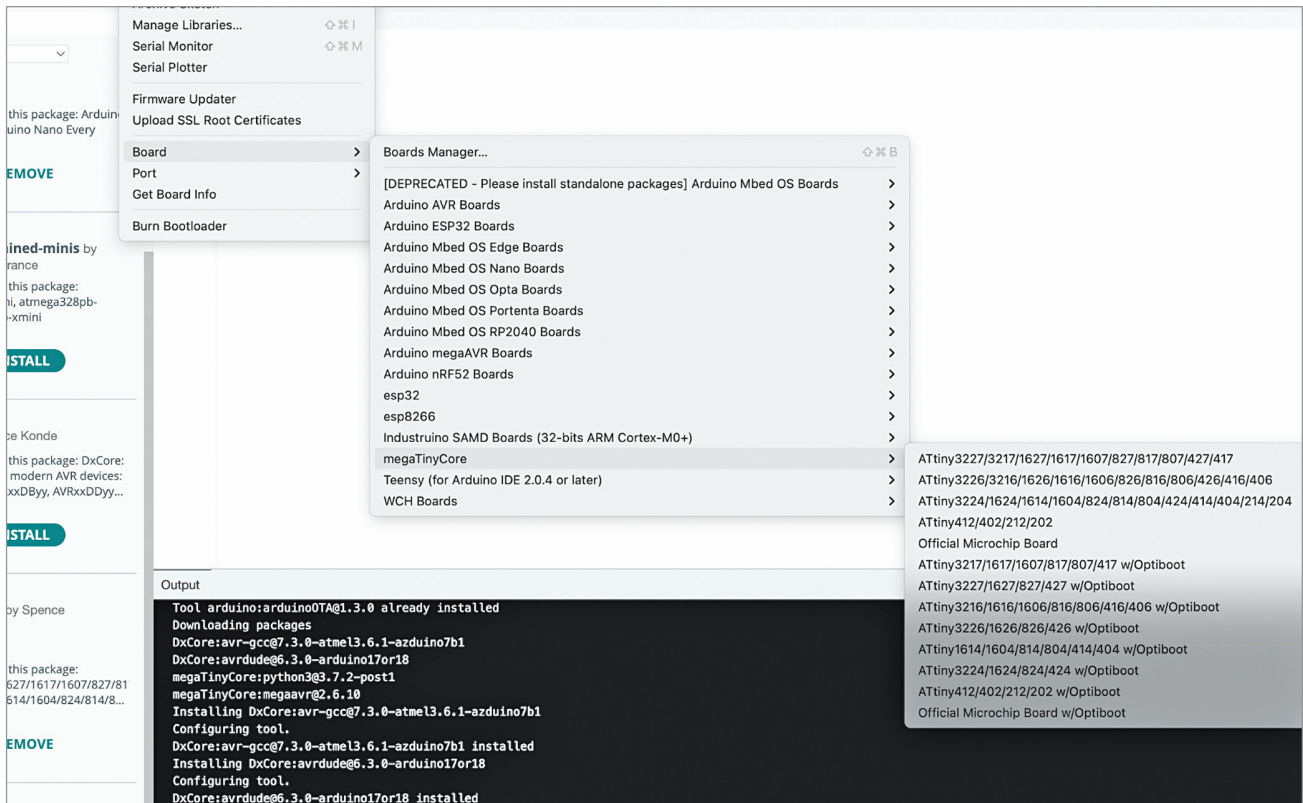


Fig. 4: Selecting the ATtiny412 MCU chip in the Arduino IDE

- Code reference: if (pressTime == 10000) in the main loop.
- Function: Decrease volume
- How it works: Holding the button for 30 seconds decreases the volume (down to a minimum of 0). The volume is adjusted using ``RDA_setVolume()`` and saved to EEPROM. Volume continues to decrease incrementally while the button remains pressed beyond the 30-second mark.

- Code reference: if (pressTime == 30000) in the main loop.
- **How to modify the switch functions.** To customise switch functionality or adjust the timing thresholds, the following code modifications may be made:
- Change timing thresholds: Edit the ``pressTime`` values in the loop to define different durations (in milliseconds).

For example, to change the seek

down function from 2 seconds to 3 seconds,

replace: if (pressTime == 2000)
with: if (pressTime == 3000)

Flashing firmware

To program the ATtiny412 MCU, either a UPDI programmer or an Arduino Uno configured as a programmer may be used. In this system, the Arduino Uno was used.

As the ATtiny412 is a surface-

mount device (SMD), two options exist for accessing its pins during programming:

1. Use an ATtiny SMD breakout board
2. Solder wires directly to the IC pins (SMD soldering method)

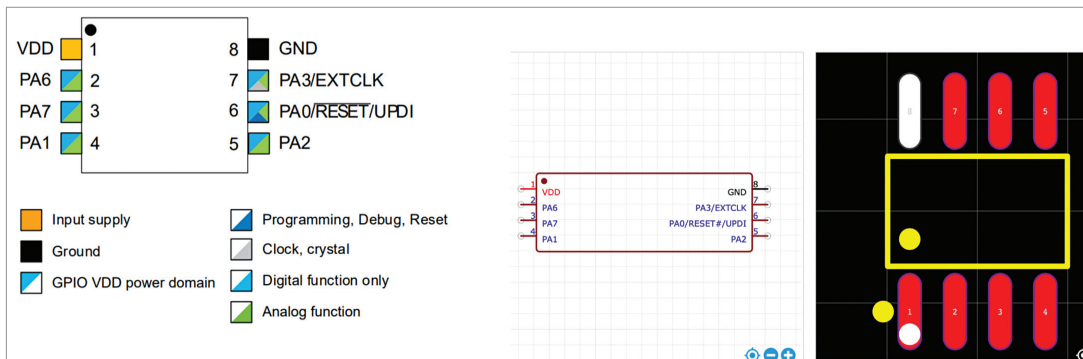


Fig. 5: ATtiny412 pinout and EDA footprint